Overview of Testing Approach

What I Want as a QA:

- a chronological ordering of the work the team has done (something similar to a `git log`)
  - **Why?** To easily answer questions like “When was Feature X completed?“
- a chronological ordering of the work that has been released (also something similar to a `git log`)
  - **Why?** To easily answer questions like “When was Feature X released?“
- a mapping between user stories and corresponding test cases and documented testing
  - **Why?** To easily answer questions like “How did we test Feature X?“

References:

- ISTQB
  - Syllabi for all certifications are freely avaiable
- Black Box Software Testing courses (Cem Kaner)
  - several freely available course recordings
- ISO/IEC 29119 (1-4) Software and systems engineering - Software testing
- ISO/IEC 25010 Software Quality Model
  - An Exploration of the ISO/IEC 25010 Software Quality Model
  - ISO 25010: Enhancing Our Software Quality Management Process
- ISO 9000 Quality Management Systems

Quality Assurance models & standards

https://ghcrajan.wordpress.com/wp-content/uploads/2013/06/cs621-unit-iii.pdf

- ISO/IEC 15504 - SPICE (Software Process Improvement and Capability dEtermination)
- ISO 9000 Series of Standards - What is ISO 9000? | ASQ
- Capability Maturity Model / Test Maturity Model
- Bohem’s Quality Model
- McCall’s Quality Model
- The Demming Cycle (PDCA)

Key Concepts

- Verification & Validation
  - Verification and validation in software development
  - System Resilience Part 6: Verification and Validation
  - Verification Vs Validation - GeeksforGeeks
  - Verification vs. Validation in Software Testing Processes
  - 5.3 Product Verification
- Quality Assurance vs. Testing
- The Role of QA in various SDLC phases
- Software Testing Life Cycle
- Test Pyramid
  - Test levels
    (unit/integration/system/E2E/user acceptance)
- Shift-left Testing
  - Continuous Testing (part of CI/CD)
- Traceability
  - Guide to Traceability in Testing | QA testing software - Kualitee
  - Importance of Traceability Matrix in Testing | BrowserStack
- Test Policy
  - Study business needs and objectives
  - Defines what should be accomplished by testing, but not how testing is to be performed
  - Define test goals traceable to business needs and objectives
  - Define test perfomance indicators
    - Guide to the top 20 QA metrics that matter - TestRail
    - What are the KPIs of Software Testing and QA Metrics?
  - Revisit and revise the test goals as appropriate
- Test Strategy
  - perform general product risk assessment to identify overtly critical testing areas
  - Define the test strategy providing clear linkage to the defined test policy and goals
  - identify and defines the appropriate test levels, and for each level define
    - objectives
    - responsibilities
    - main tasks

    - general entry/exit criteria
- Test Plan
    - How To Create A Test Plan (Steps, Examples, & Template) - TestRail
- Test types
  (manual/automated, functional/non-functional)
    - Types of Software Testing - GeeksforGeeks
    - Different Types of Testing in Software | BrowserStack
- Test techniques
  (Structure-based/white box, Requirements-based/black box, Experience-based/exploratory)
    - Software Testing Techniques - GeeksforGeeks
    - https://wildart.github.io/MISG5020/standards/ISO-IEC-IEEE-29119-4.pdf
- Test Entry and Exit Criteria
    - Entry and Exit Criteria in Software Testing | BrowserStack
- Test Coverage
    - Test Coverage Techniques Every Tester Must Know | BrowserStack
    - A Detailed Guide on Test Coverage
- Test Estimation
    - Software Testing - Estimation Techniques - GeeksforGeeks
- Work Breakdown Structure
    - Work Breakdown Structure - Software Engineering - GeeksforGeeks

The role of QA in the various SDLC phases

- The Role of QA in the Software Development Life Cycle

1. Planning
    a. be an advocate for and expert on user experience
2. Design
    a. Advocate for testability
3. Implementation
    a. Advocate for testability
4. Testing
5. Deployment
6. Maintenance

Notes:

- I noticed that the five TMMi Level 2 Process Areas (Policy & Strategy, Planning, Monitoring & Control, Design & Execution, Environment) map fairly well onto the ISO/IEC/IEEE 29119-2 Test Processes (Organizational, Management, Monitoring & Control, Completion, Dynamic).
- Re SDLC: a good QA system is one that has clear links between the outputs of the Planning & Design phases. and the Implementation phases
- Donald Firesmith's article A Taxonomy of Testing has an interesting way of organizing the dizzying number of types of testing:

> I initially created 12 general types of testing, which eventually grew to 16. One of my colleagues, however, pointed out that a better way would be to first organize them by the way they primarily answer one of the following standard sets of questions, commonly known as the 5Ws (*who*, *what*, *when*, *where*, and *why*), and an 2Hs (*how* and *how well*). This in turn led to the development of the following questions, which formed the foundation of my taxonomy for organizing testing types:
>
> ***What** are we testing?*
>
> ***When** are we testing?*
>
> ***Why** are we testing?*
>
> ***Who** is performing the testing?*
>
> ***Where** is the testing taking place?*
>
> ***How** are we testing?*
>
> ***How well** are the objects-under-test functioning?*

Connections:

- Verification & Validation ⟷ Shift-left testing ⟷ SDLC
  - Shifting testing earlier in the development lifecycle means including testers in the Verification process ("Are we building the right thing?"), whereas testers usually primarily operate within the Validation process ("Did we build the thing right?")
- Shift-left testing ⟷ SDLC ⟷ Continuous Testing ⟷ Test automation & Test Pyramid

- Verification & Validation ⟷ SDLC
  - SDLC timing: Verification activities precede Validation activities because a software product has to be **developed and built** before it can be executed and thus Validated. Verification focuses on the software product itself while Validation focuses on the user's experience with the software product
- Traceability ⟷ Shift Left
  - Give developers feedback on the traceability of their code as soon as possible in the development process

Questions to answer:

- ~~What makes up a Test Plan?~~
- What is the Software Testing Life Cycle?
- ~~What is the Test Pyramid~~
- ~~What is shift-left?~~
- ~~What is the objective of testing?~~
  - ~~Verification vs. Validation~~
  - ~~Quality Assurance vs. Testing~~
- ~~What is the difference between testing and Quality Assurance?~~